

1 Reduction Rules

Note on terminology: it is assumed that variable names are automatically renamed to avoid capture. For example, in the reduction rule for β -reduction, we assume that the variable x is not free in N .

2 Grammar

The following rules are adapted from Barendregt's *Lambda Calculus with Types* (1992) (Page 92).

$$\begin{aligned} \langle \text{Expr} \rangle ::= & x \\ & | \lambda \langle \text{Id} \rangle^{\langle \text{Expr} \rangle} . \langle \text{Expr} \rangle \\ & | \text{let } \langle \text{Id} \rangle^{\langle \text{Expr} \rangle} := \langle \text{Expr} \rangle \text{ in } \langle \text{Expr} \rangle \\ & | \langle \text{Expr} \rangle \langle \text{Expr} \rangle \\ & | f_n \end{aligned}$$

3 Typing Rules

4 Why are predicativity results important

according to <https://inria.hal.science/inria-00076023/file/RR-0531.pdf>,

5 Proof of Normalization

Definition 5.1. The notation

$$\Gamma \vdash \mathcal{WT} \{e\}$$

will be used to denote that an expression e is **well-typed** in a context Γ .

An expression is well-typed if it has a well-typed type label and all of its sub-expressions are well-typed. In addition, all Type literals are well-typed.

Writing these as syntactic rules in the style of

Theorem 5.1. *As defined in https://wiki.haskell.org/Weak_head_normal_form and <https://j-hui.com/pages/normal-forms/#whnf>, every type expression is in weak head normal form.*

Proof. The statement

□

6 Predicativity and Semantics

According to <https://math.stanford.edu/~feferman/papers/predicativity.pdf>, predicativity is kind of a murky concept. I will formally define predicativity as the inability to construct the semantics of a system. I want to claim that with pure type systems, this only happens via a mechanism I describe in the paragraph below.

In particular, a type in a type system (this covers sorts since sorts are types) is impredicative if the natural semantics allow you to prove that that universe is at least as large as its power set. i.e. you have an injection from the $(U \rightarrow T) \hookrightarrow U$, where $|T| \geq 2$. This is a contradiction by diagonalization, which we discussed in class.

Two expressions are in the same level iff they share some n -type. It is trivial to prove that this is an equivalence relation for typed expressions.

The
normalizing

7 Hurkens' Paradox

This is an adaptation of Girard's paradox.

Introduced in Hurkens 1995, this paradox is a proof of False within the pure type system of U^- . Clearly, any $* : *$ system also contains this paradox because this is just a stronger system. In particular, it has the additional rule that all sorts are interchangeable.

I think that the original proof is rather frugal on intuition, so I will attempt to give a more intuitive explanation of the paradox. In particular, I will focus on the process of encoding the paradox within our system.

7.1 Paradox in Graph Theory

Hurkens originally expressed the high-level paradox in terms of naive set theory, but I will translate this to graph semantics for a few reasons:

- Graph semantics are more intuitive to me.
- The syntax of the proof is two-typed, and I think NST fails to convincingly adhere to this two-typed structure.
- Graph semantics allow me to draw fun visuals for the presentation.

I am also writing the proof steps more explicitly. In the original article, Hurkens writes the proof in a way that assumes a lot of prior knowledge on behalf of the reader. I will rewrite the proof in a way that is more explicit and hopefully easier to follow, at the cost of being more verbose. Let $\mathcal{U}$ be some set of vertices. Let σ be the edge predicate on $\mathcal{U}$, but inverted. In other words, $\sigma(u, v)$ is true iff there is an edge from v to u . We also allow currying, so $\sigma(u)$ is the set of all vertices v such that there is an edge from v to u . In other words,

$\sigma(u)$ is the set of all direct predecessors of u . Let τ be a function from sets of vertices to vertices.

As stated in the original paper, we say that σ , τ , and $\mathcal{U}$ are paradoxical if the following is true:

$$\forall X \subseteq \mathcal{U} : \sigma(\tau(X)) = \{\tau(\sigma(x)) \mid x \in X\}$$

Using graph semantics, we can also say

$$\forall (X \subseteq \mathcal{U}, k \in \mathcal{U}) : \sigma(\tau(X), k) \leftrightarrow (\exists x \in X : k = \tau(\sigma(x)))$$

Furthermore, as noted by Hurkens, for most of the proof, we will only use one direction:

$$\forall X \subseteq \mathcal{U} : \sigma(\tau(X)) \subseteq \{\tau(\sigma(x)) \mid x \in X\}$$

I will explicitly call out when we use the other direction. Note that $\sigma \circ \tau$ is a function from sets of vertices to sets of vertices, and $\tau \circ \sigma$ is a function from vertices to vertices. The above condition says that $\sigma \circ \tau$ is effectively the same as $\tau \circ \sigma$, but lifted into the power set.

As you may have noticed, calling this property *paradoxical* begs the question of how it leads to a paradox.

First, let's prove some preliminary lemmas about the structure of $(\mathcal{U}, \sigma, \tau)$.

Lemma 7.1. $\tau \circ \sigma$ is an endomorphism of $\mathcal{U}$.

Proof. Suppose $\sigma(u, v)$. Our goal is to prove $\sigma(\tau(\sigma(u)), \tau(\sigma(v)))$. By predicate encoding, this is equivalent to $\tau(\sigma(v)) \in \sigma(\tau(\sigma(u)))$. By paradoxicality on $\sigma(\tau(\sigma(u)))$, our goal is equivalent to $\tau(\sigma(v)) \in \{\tau(\sigma(w)) \mid w \in \sigma(u)\}$. We assumed that $\sigma(u, v)$, so by predicate encoding, $v \in \sigma(u)$. By set theory, (matching v with w), $\tau(\sigma(v)) \in \{\tau(\sigma(w)) \mid w \in \sigma(u)\}$, and our goal is satisfied.

Here is the proof written in a deductive style.

$$\begin{array}{ll} \text{Suppose} & \sigma(u, v) \\ \text{By predicate encoding} & v \in \sigma(u) \\ \text{By set theory (matching } v \text{ with } w) & \tau(\sigma(v)) \in \{\tau(\sigma(w)) \mid w \in \sigma(u)\} \\ \text{By paradoxicality} & \tau(\sigma(v)) \in \sigma(\tau(\sigma(u))) \\ \text{By predicate encoding} & \sigma(\tau(\sigma(u)), \tau(\sigma(v))) \end{array}$$

□

Definition 7.1 (Inductive Predicate). A subset P of $\mathcal{U}$ is *inductive* if for every vertex u , if every predecessor v of u is in P , then u is in P as well. Written formally, P is inductive if

$$\forall u : [\forall v : \sigma(u, v) \rightarrow v \in P] \rightarrow u \in P$$

Lemma 7.2. For any inductive subset P , the set $Q = \{v \mid \tau(\sigma(v)) \in P\}$ is inductive.

Proof. Suppose there is some vertex u such that every predecessor v of u is in Q . Our goal is to show that u is in Q as well, or equivalently, $\tau(\sigma(u)) \in P$. By definition of Q , this is equivalent to saying that for every predecessor v of u , $\tau(\sigma(v)) \in P$. Let w be an arbitrary predecessor of $\tau(\sigma(u))$. Because P is inductive, it is sufficient to show that w is in P . By predicate encoding, $w \in \sigma(\tau(\sigma(u)))$. By paradoxicality, $w \in \{\tau(\sigma(x)) \mid x \in \sigma(u)\}$. By set theory, there is some x such that $w = \tau(\sigma(x))$ and $x \in \sigma(u)$. Because x is a predecessor of u , we have that $\tau(\sigma(x))$ is in P . Because $w = \tau(\sigma(x))$, we have that w is in P . Our goal is satisfied.

Here is the proof written in a deductive style.

Suppose	$\forall v : v \in \sigma(u) \rightarrow v \in Q$
Suppose	$w \in \sigma(\tau(\sigma(u)))$
By paradoxicality	$w \in \{\tau(\sigma(x)) \mid x \in \sigma(u)\}$
By set theory	$w = \tau(\sigma(x)) \wedge x \in \sigma(u)$
By assumption 1	$x \in Q$
By definition of Q	$\tau(\sigma(x)) \in P$
By equality	$w \in P$
Generalizing w	$\forall w : w \in \sigma(\tau(\sigma(u))) \rightarrow w \in P$
By inductiveness	$\tau(\sigma(u)) \in P$
By definition of Q	$u \in Q$

We can avoid equality by using subsetting logic (when we formalize, we actually find that paradoxicality encodes the set construction as the intersection of all supersets).

Suppose	$\forall v : v \in \sigma(u) \rightarrow v \in Q$
Suppose	$w \in \sigma(\tau(\sigma(u)))$
By paradoxicality	$w \in \{\tau(\sigma(x)) \mid x \in \sigma(u)\}$
Suppose	$x \in \sigma(u)$
By assumption 1	$x \in Q$
By definition of Q	$\tau(\sigma(x)) \in P$
Generalizing x	$\forall x : x \in \sigma(u) \rightarrow \tau(\sigma(x)) \in P$
By set theory	$\{\tau(\sigma(x)) \mid x \in \sigma(u)\} \subseteq P$
By transitivity	$w \in P$
Generalizing w	$\forall w : w \in \sigma(\tau(\sigma(u))) \rightarrow w \in P$
By inductiveness	$\tau(\sigma(u)) \in P$
By definition of Q	$u \in Q$

□

Definition 7.2 (Well-foundedness). A vertex u is well-founded if for every inductive predicate P , u is in P .

We additionally define Ω to be the vertex $\tau\{u \in \mathcal{U} \mid u \text{ is well-founded}\}$.

Lemma 7.3. *Ω is well-founded.*

Proof. Let P be some inductive predicate. Let k be some predecessor of Ω . We have that $k \in \sigma(\Omega)$. By paradoxicality,

$$k \in \{\tau(\sigma(x)) \mid x \text{ is well-founded}\}$$

Structurally, we can say $k = \tau(\sigma(w))$ where w is well-founded. Because w is well-founded, it is in every inductive predicate. Let P be some inductive predicate. Consider the set $Q = \{y \mid \tau(\sigma(y)) \in P\}$. By the previous lemma, Q is inductive. Because w is well-founded, it is in Q , and so $\tau(\sigma(w))$ is in P . By definition of k , k is in P as well. Generalizing over k , every predecessor of Ω is in P . By inductiveness, Ω is in P as well. Generalizing over P , Ω is in every inductive predicate. By definition, Ω is well-founded.

This is already written deductively, but we can avoid equality as follows:

Suppose	P is inductive
Suppose	$k \in \sigma(\Omega)$
By paradoxicality	$k \in \{\tau(\sigma(x)) \mid x \text{ is well-founded}\}$
Suppose	w is well-founded
By previous lemma	$\{x \mid \tau(\sigma(x)) \in P\}$ is inductive
By definition of well-founded	$w \in \{x \mid \tau(\sigma(x)) \in P\}$
By set theory	$\tau(\sigma(w)) \in P$
Generalizing w	$\forall w : w \text{ is well-founded} \rightarrow \tau(\sigma(w)) \in P$
By set theory	$\{\tau(\sigma(w)) \mid w \text{ is well-founded}\} \subseteq P$
By transitivity	$k \in P$
Generalizing k	$\forall k : k \in \sigma(\Omega) \rightarrow k \in P$
By inductivity	$\Omega \in P$
Generalizing P	Ω is well-founded

□

Lemma 7.4. *The set $\{y \mid \tau(\sigma(y)) \notin \sigma(y)\}$ is inductive*

Proof. Let x be some vertex where every predecessor y satisfies $\tau(\sigma(y)) \notin \sigma(y)$. Suppose $\tau(\sigma(x)) \in \sigma(x)$. By endomorphism, $\tau(\sigma(\tau(\sigma(x)))) \in \sigma(\tau(\sigma(x)))$

Our assumption also gives us that $\tau(\sigma(x))$ is a predecessor of x , and thus we can derive an absurdity from $\tau(\sigma(\tau(\sigma(x)))) \in \sigma(\tau(\sigma(x)))$. □

Lemma 7.5. *False*

Proof. We previously showed that Ω is well founded. Remember that by paradoxicality, $\sigma(\Omega) = \{\tau(\sigma(x)) \mid x \text{ is well-founded}\}$. Because Ω is well-founded, $\tau(\sigma(\Omega)) \in \sigma(\Omega)$. Because Ω is well-founded, $\Omega \in \{y \mid \tau(\sigma(y)) \notin \sigma(y)\}$. By set theory, $\tau(\sigma(\Omega)) \notin \sigma(\Omega)$. We can derive any absurdity we desire :)

□

7.2 Formalization

Intuitively, we know that in system U^- we can construct

- There are level-0 objects that depend on level-0 objects.
- There are level-1 objects that depend on level-0 objects.
- There are level-1 objects that depend on level-1 objects.
- There are level-2 objects that depend on level-1 objects.

level 2 and 3 are thus a bit barren. Kind of like the “attic” of the type system. level 2 contains high-level Π types plus $\Box$, and level 3 is just Δ .

7.2.1 Encodings

Our vertices will be level-1 objects. Our edge predicate will also be a level-1 object (a function that takes two level-1 objects and returns a proposition, which is a level-1 object). The τ function is also a level-1 object. Subsets of $\mathcal{U}$ are encoded as level-1 predicates, which are also level-1 objects.

In the original article by Hurkens, he introduces the following terms:

$$\mathcal{U} \equiv \Pi \mathcal{X} : \square. ((\mathcal{X} \rightarrow *) \rightarrow \mathcal{X}) \rightarrow \mathcal{X} \rightarrow * \quad (1)$$

$$\tau \equiv \lambda X : \mathcal{U} \rightarrow *. \quad (2)$$

$$\lambda \mathcal{A} : \square.$$

$$\lambda c : (\mathcal{A} \rightarrow *) \rightarrow \mathcal{A}.$$

$$\lambda a : \mathcal{A}.$$

$$\Pi P : \mathcal{A} \rightarrow *.$$

$$\Pi x : \mathcal{U}.$$

$$[X \ x \rightarrow P(c(x \ \mathcal{A} \ c))] \rightarrow$$

$$P \ a$$

$$\sigma \equiv \lambda x : \mathcal{U}. \quad (3)$$

$$x \ \mathcal{U} \ \tau$$

Applying typing rules, we get:

$$\Gamma :: [x : \mathcal{U}] \vdash (x \ \mathcal{U} \ \tau) : \mathcal{U} \rightarrow * \quad (4)$$

$$\Gamma \vdash \sigma : \Pi x : \mathcal{U}. \mathcal{U} \rightarrow * \quad (5)$$

To verify that this is paradoxical, we want to show that

$$\begin{aligned} \sigma(\tau \ X) &\equiv_{\beta} \lambda u : \mathcal{U}. \\ &\quad \Pi P : \mathcal{U} \rightarrow *. \\ &\quad [\Pi x : \mathcal{U}. X \ x \rightarrow P(\tau(x \ \mathcal{U} \ \tau))] \rightarrow \\ &\quad P \ u \end{aligned}$$

$$\sigma(\tau \ X) \equiv_{\delta}$$

$$(\lambda(x : \mathcal{U}). x \ \mathcal{U} \ \tau)(\tau \ X) \equiv_{\beta}$$

$$(\tau \ X) \ \mathcal{U} \ \tau \equiv$$

$$\tau \ X \ \mathcal{U} \ \tau \equiv_{\delta, \beta[X/X][\mathcal{A}/\mathcal{U}][c/\tau]}$$

$$\lambda a : \mathcal{U}. \Pi P : \mathcal{U} \rightarrow *. \Pi x : \mathcal{U}. ([X \ x \rightarrow P(\tau(x \ \mathcal{U} \ \tau))] \rightarrow P \ a) \equiv_{\alpha[a/u]}$$

$$\lambda u : \mathcal{U}. \Pi P : \mathcal{U} \rightarrow *. \Pi x : \mathcal{U}. ([X \ x \rightarrow P(\tau(x \ \mathcal{U} \ \tau))] \rightarrow P \ u)$$

We can express the inductive predicate.

$$\mathbb{I} \equiv \lambda P : \mathcal{U} \rightarrow *. \Pi u : \mathcal{U}. [\Pi v : \mathcal{U}. \sigma \ u \ v \rightarrow P \ v] \rightarrow P \ u$$

$$\mathbb{I} : \Pi P : \mathcal{U} \rightarrow *. *$$

$$\Pi P : \mathcal{U} \rightarrow *. * : \square$$

Thus, $\mathbb{I}$ is a level-1 object.

We can also express the well-founded predicate.

$$W \equiv \lambda u : \mathcal{U}. \Pi P : \mathcal{U} \rightarrow *. \mathbb{I} P \rightarrow P u$$

Then, we can define Ω as

$$\Omega \equiv \tau(\lambda u : \mathcal{U}. W(u))$$

We can prove a beta-reduction:

$$\begin{aligned} \sigma \Omega k &\equiv_{\delta} \\ \sigma(\tau(\lambda u : \mathcal{U}. W u)) k &\equiv_{\text{paradoxicality}} \\ (\lambda u : \mathcal{U}. \Pi P : \mathcal{U} \rightarrow *. [\Pi x : \mathcal{U}. (\lambda u : \mathcal{U}. W u) x \rightarrow P(\tau(\sigma x))] \rightarrow P u) k &\equiv_{\beta} \\ \Pi P : \mathcal{U} \rightarrow *. [\Pi x : \mathcal{U}. (\lambda u : \mathcal{U}. W u) x \rightarrow P(\tau(\sigma x))] \rightarrow P k &\equiv_{\beta} \\ \Pi P : \mathcal{U} \rightarrow *. [\Pi x : \mathcal{U}. W x \rightarrow P(\tau(\sigma x))] \rightarrow P k & \end{aligned}$$

Let's prove that k is in every inductive set.

$$\begin{aligned} k &: \mathcal{U} \\ p_0 &: \sigma \Omega k \\ \Pi P &: \mathcal{U} \rightarrow * \\ p_1 &: \mathbb{I} P \end{aligned}$$

We show that $\{x \mid \tau(\sigma x) \in P\}$ is inductive.

$$Q \equiv \lambda x : \mathcal{U}. P(\tau(\sigma x))$$

TODO endomorphism should just beta reduce

$$\begin{aligned} p_2 &: \Pi u : \mathcal{U}. \Pi v : \mathcal{U}. \sigma u v \rightarrow \sigma(\tau(\sigma u))(\tau(\sigma v)) \\ p_3 &\equiv \lambda u : \mathcal{U}. \lambda p_4 : (\Pi v : \mathcal{U}. \sigma u v \rightarrow P(\tau(\sigma(v)))) . (\text{something } Pto \end{aligned}$$

Intuitively, this says k being a predecessor of Ω is equivalent to k being in every vertex set P that contains every $\tau(\sigma x)$ where x is well-founded.

$$\begin{aligned} Wk &\equiv_{\delta} \\ (\lambda u : \mathcal{U}. \Pi P : \mathcal{U} \rightarrow *. \mathbb{I} P \rightarrow P u) k &\equiv_{\beta} \\ \Pi P : \mathcal{U} \rightarrow *. \mathbb{I} P \rightarrow P k &\equiv_{\delta} \\ \Pi P : \mathcal{U} \rightarrow *. (\lambda P : \mathcal{U} \rightarrow *. \Pi u : \mathcal{U}. [\Pi v : \mathcal{U}. \sigma u v \rightarrow P v] \rightarrow P u) P \rightarrow P k &\equiv_{\delta} \\ \Pi P : \mathcal{U} \rightarrow *. (\Pi u : \mathcal{U}. [\Pi v : \mathcal{U}. \sigma u v \rightarrow P v] \rightarrow P u) \rightarrow P k &\equiv_{\delta} \end{aligned}$$

Every predecessor of Ω is well-founded.

$$\begin{aligned} p_0 &: \Pi k : \mathcal{U}. \sigma \Omega k \rightarrow Wk \\ p_0 &\equiv \lambda k : \mathcal{U}. \lambda p_1 : \sigma \Omega k. \rightarrow W(k) \end{aligned}$$

Ω is well founded.

$$p_0 \equiv \lambda P : (\mathcal{U} \rightarrow *). \lambda (p_1 : \mathbb{I}P). (\textit{something of type } P\Omega)$$

Recall that a classical operator (function that takes a type and gives an element or an ex-falso style refutation) assumes that you can have values that depend on types. i.e. the value level is impredicative. According to section 4.5.3 of <https://inria.hal.science/inria-00075471/file/RR-1088.pdf>, classical logic forces proof irrelevance.

8 Proof that typing is finite time

see <https://groups.seas.harvard.edu/courses/cs152/2014sp/lectures/lec10-types.pdf>